



Slippery skin

Fish make their bodies slimy with mucus, but dolphins reduce drag by scattering skin cells as they speed through the sea. Dolphin skin also ripples with the water, making dolphins a hundred times more slippery than a human swimmer.

Self-repair

This lion's face was cut during a fight. Skin often gets damaged—either during skirmishes or by accident.

Once this happens, the injury soon begins to heal. A blood clot forms over the damaged area, and skin cells start to multiply, forming a scar. Unlike normal skin, scar tissue is harder and less flexible, with fewer blood vessels and nerves.



Scar tissue lacks fur



Practical protection

Vultures may not look beautiful, but having bare skin on their heads and necks is ideal for their scavenging way of life. It allows the birds to put their heads inside a carcass without getting their feathers covered in blood. Like all vertebrates, a vulture's skin has two main layers. The deepest one, called the dermis, contains blood vessels and nerves. The layer above it, called the epidermis, produces a constant supply of dead cells containing a protein called keratin. These dead cells form a waterproof barrier on the skin's surface.

Chemical factory

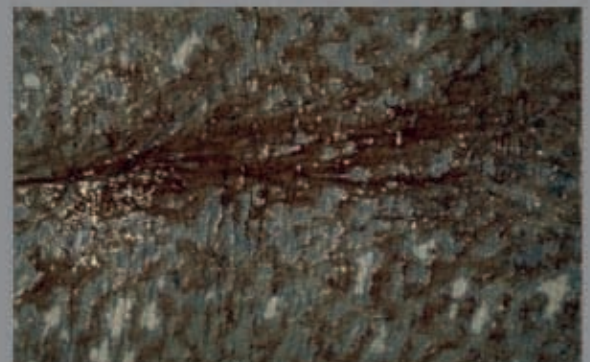
Many animals produce different chemical substances in their skin. Some of the chemicals help to defend animals against predators. Other secretions help to protect an animal's body against the effects of their environment.

Wartlike bumps called paratoid glands ooze poisonous chemicals



Poisonous surface

The fire salamander secretes poison onto its skin to deter predators. The poisons are made in the paratoid glands, which are the two rows of bumps along the salamander's back and the pair of bumps behind its eyes.



Natural sunscreen

Hippopotamus skin produces droplets of oil that screen the ultraviolet rays in sunshine. This oil also helps to fight waterborne bacteria that can infect a hippopotamus's wounds when wallowing in water.



Sleeping bag

At sunset, some parrotfish hide away in rock crevices and secrete a mucus cocoon from their skin. This jellylike blanket probably masks their scent, making it harder for predators and parasites to find the fish as they sleep.